

The Hidden-37 Triangle Theorem

Cyclic decimal rotations, Heron's formula, and irrational triangle area

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Main result. If the cyclic rotations of a three-digit decimal block form a positive, nondegenerate triangle, then its area and all three altitudes are irrational. In every non-equilateral case, 37 occurs exactly once in 16 times the squared area.

Abstract

For digits a, b, c , let A, B, C be the integers formed by cyclically rotating abc . If these numbers are positive and satisfy the strict triangle inequalities, then the corresponding triangle has irrational area and irrational altitudes. More exactly, for non-equal digits, $v_{37}(16D^2)=1$, where D is the area. Exact enumeration of all 1,000 blocks finds 534 admissible inputs: 525 non-equilateral and 9 equilateral, with no counterexamples.

1. Statement

Define

$$A=100a+10b+c, \quad B=100b+10c+a, \quad C=100c+10a+b.$$

Theorem. Suppose A, B, C are positive and form a nondegenerate triangle with area D. Then D and all three altitudes are irrational. If a,b,c are not all equal, $v_{37}(16D^2)=1$. If $a=b=c$, then $v_{37}(16D^2)=4$.

2. Proof

Let $P=A+B+C$. Then

$$P=111(a+b+c)=3 \times 37 \times (a+b+c).$$

A valid triangle has $1 \leq a+b+c \leq 27$, so $v_{37}(P)=1$. Since 1000 is congruent to 1 modulo 37, cyclic rotation is multiplication by 10 modulo 37: B is congruent to 10A and C to 10B. Consequently, if one side is divisible by 37, all three are.

All rotations have the same digit sum, so their pairwise differences are divisible by 9. If the sides were also divisible by 37, the differences would be divisible by 333. When the digits are not all equal, the rotations are distinct. Sort them $x=333$ and $z-y \geq 333$. Because $z \leq 999$, we obtain $x \leq 333$, and therefore $x+y \leq 333+y \leq z$, contradicting the strict triangle inequality. Thus no side of a non-equilateral admissible triangle is divisible by 37.

Heron's integral identity is

$$16D^2=P(P-2A)(P-2B)(P-2C).$$

Modulo 37, $P-2A$ is congruent to $-2A$ and is nonzero; similarly for B and C. Therefore only P contributes a factor of 37, and it contributes exactly one. Hence $v_{37}(16D^2)=1$. A perfect square has even valuation at every prime, so $16D^2$ is not a square and D is irrational. Every altitude is $2D$ divided by an integer side, so every altitude is irrational.

If $a=b=c=d$, then $A=B=C=111d$ and $16D^2=3(111d)^4$. Its 37-adic valuation is 4, but D and the altitudes remain irrational because they are nonzero rational multiples of $\sqrt{3}$.

3. Example

For 123, the sides are 123, 231, and 312. Exact calculation gives

$$D=126 \sqrt{9435}=126 \sqrt{37 \times 255},$$

$$h_{312}=(21/26) \sqrt{37 \times 255}.$$

4. Exact computational check

Check	Result
Blocks examined	1,000
Triangle-producing	534
Non-equilateral	525
Equilateral	9
Counterexamples	0

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from math import sqrt
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fail=[]; valid=non_eq=0
for n in range(1000):
    a,b,c=n//100,(n//10)%10,n%10
    A=100*a+10*b+c; B=100*b+10*c+a; C=100*c+10*a+b
    P=A+B+C
    if min(A,B,C)<=0 or 2*max(A,B,C)>=P: continue
    valid += 1
    R=P*(P-2*A)*(P-2*B)*(P-2*C)
    if isqrt(R)**2==R: fail.append((n,"square"))
    if not(a==b==c):
        non_eq += 1
        if R%37 or R%(37**2)==0: fail.append((n,"37 rule"))
assert (valid,non_eq,fail)==(534,525,[])

```

5. Novelty assessment

A broad search completed in September 2026 found no earlier source stating the complete theorem or an evidently equivalent result. The theorem therefore appears plausibly original as a combined, explicitly quantified statement, especially the exact valuation $v_{37}(16D^2)=1$.

The individual ingredients are established: Heron's formula, divisibility facts involving 111 and 37, cyclic rotation of three-digit multiples of 37, and integer-sided triangles with irrational area. The apparent novelty is their synthesis with the triangle inequality.

Recommended claim: “We are not aware of a previous explicit observation that every non-equilateral triangle whose integer side lengths are the three cyclic rotations of one three-digit decimal numeral satisfies $v_{37}(16D^2)=1$, and consequently has irrational area and irrational altitudes.”

This is evidence of novelty, not proof of worldwide priority. An unindexed or unpublished source could contain the same observation.

6. Base-b perspective

The decimal result is part of a broader mechanism. For three base-b rotations, their sum contains the factor b^2+b+1 . A prime dividing this factor may similarly enter the perimeter while remaining absent from the other Heron factors. The decimal case is $b=10$, where $10^2+10+1=111=3 \times 37$.

Conclusion

Decimal rotation forces one factor of 37 into the perimeter; modular rotation controls the remaining Heron factors; and the triangle inequality excludes the only possible extra divisibility in non-equilateral cases. The result is an exact arithmetic obstruction to rational area.